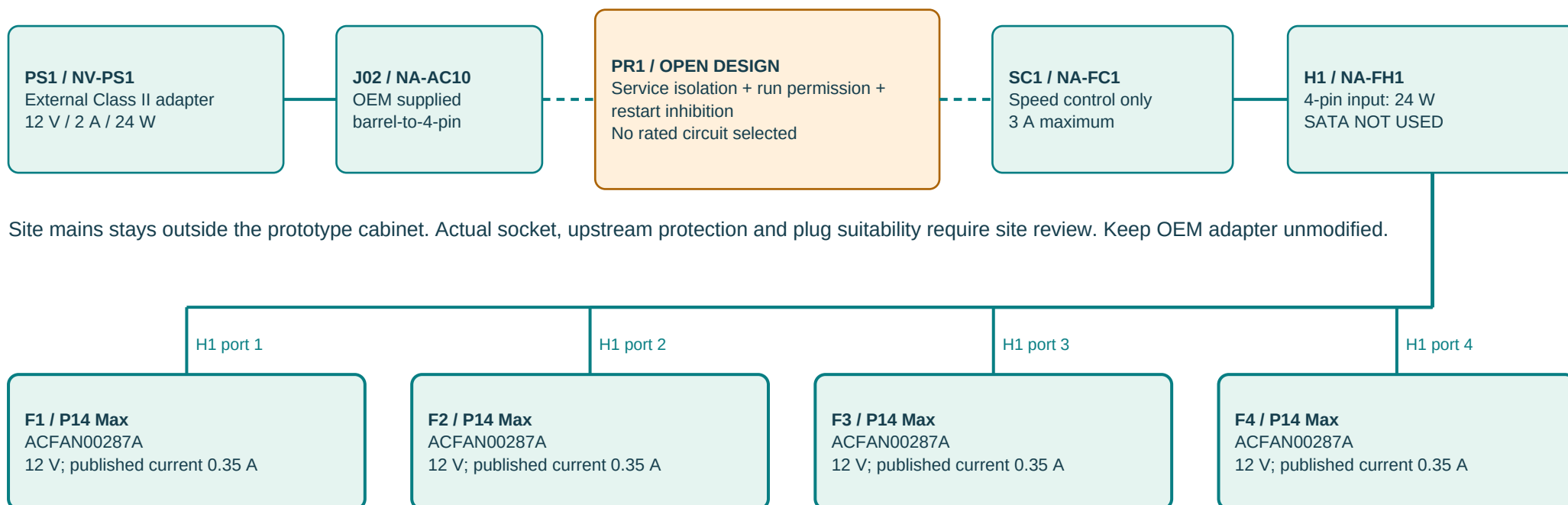


One power path, four fan branches

OEM reference connectors are named; amber functions are unselected. No bridge across PR1 is permitted.



Site mains stays outside the prototype cabinet. Actual socket, upstream protection and plug suitability require site review. Keep OEM adapter unmodified.

OEM interfaces: SC1 output to H1 4-pin input using compatible OEM cable; H1 ports1–4 to individual fans. Lines represent connector-level power/control bundles, NOT conductor routing or plug orientation. PR1 insertion would require a designed compatible harness; no OEM cable should be cut to implement this sketch.

Only F1 RPM is forwarded upstream. Other port LEDs are local indications, not independent safety outputs. Do not tie tach outputs together. A lamp or zero PWM does not prove isolation or standstill.

Release hold: the pictured OEM chain alone does not implement the required STOP / deliberate RESET / fresh START behavior. PR1, cable entry, enclosure, conductor protection and all physical connections remain unreleased.

Load and thermal constraints

Catalogue limits are verified. Startup multipliers and auxiliary loads below are illustrative assumptions.

Path item	Published limit / value	Engineering interpretation
Four P14 Max fans	4 x 0.35 A = 1.40 A; 16.8 W at12 V	Published fan current; not measured startup or guaranteed maximum draw
NV-PS1 external adapter	12 V; 2 A /24 W	Governs current together with selected hub input
NA-FC1 speed controller	3 A /36 W maximum	Rating is NOT its own consumption; auxiliary load unknown
NA-FH1 selected input	24 W via4-pin; 2 A at12 V	54 W SATA rating does NOT apply to this path
Temperature	Fans0–40 C; adapter0–40 C; hub-40–60 C	Controller/enclosure environment unresolved. Not whole-device outdoor approval

Assumed fan-current multiplier	Assumed auxiliary load	Total demand	Difference to2 A
1.00x	0.00 A	1.40 A	+0.60 A
1.00x	0.10 A	1.50 A	+0.50 A
1.25x	0.10 A	1.85 A	+0.15 A
1.50x	0.00 A	2.10 A	-0.10 A
2.00x	0.00 A	2.80 A	-0.80 A

Consequence: with0.10 A assumed auxiliary load, the arithmetic fan-current multiplier ceiling is1.357x. Actual startup magnitude/duration is UNKNOWN. Exceeding a continuous catalogue limit does not predict the actual trip timing. Conversely, positive arithmetic margin is not a safety pass.

Do not simply install a stronger adapter: a5 A adapter feeding the same2 A hub path still leaves2.8 A hypothetical fan demand above the hub limit. A changed power architecture requires new connectors, protection and recovery analysis. Do not parallel adapters or connect SATA in this design.

Fan pin functions and wiring limits

OEM pin function reference only. Connector view, colour mapping and the physical terminal schedule are NOT released.

P14 Max pin	OEM function	Design restriction
1	Ground (-)	Signal/power return is not protective earth. Do not infer connector view or wire colour.
2	Power (+)	12 V reference. Rated switching and conductor protection unselected.
3	Tachometer output	Do not connect multiple tach outputs together or assume a safe-standstill signal.
4	PWM control	Speed command only; no claimed protective stop or service isolation.

Voltage-drop calculation: $\Delta V = I \times 2L \times \rho / A$. At assumed copper $\rho_{20} = 0.0175 \text{ ohm mm}^2/\text{m}$, one-way 1 m, 0.5 mm² and 1.4 A: $\Delta V = 0.098 \text{ V}$. At 40 C with assumed $\alpha 0.00393/\text{K}$: 0.106 V. These are examples, NOT the OEM cable gauge or a selected wire.

Coordination input	Current status	Needed to release
Fan terminal voltage	Allowed running range UNKNOWN	Do not use published 3.9 V startup voltage as allowed running minimum
Wire / connectors	Actual conductor gauge, contact rating, resistance UNKNOWN	OEM evidence plus routing, temperature, retention and measured terminal voltage
Fuse / protection	Hub auto-reset fuse; time-current / current-limit curves UNKNOWN	Coordinate fault energy with weakest conductor/contact and rated DC device; no arbitrary 2 A fuse
Switching / enclosure	PR1 and enclosure UNSELECTED	Actual DC load/inrush ratings, compatible harness, cooling and service/isolation arrangement

Keep the logger independent: no GPIO, USB or laptop feed into the fan-power or protective path. OEM PSU Class II does not decide the complete prototype protection class or whether any conductive parts require bonding; assess the actual assembly.

Protection coverage and release evidence

Digital work defines the gap; it does not fabricate a protective relay or claim physical commissioning.

Event / hazard	OEM chain evidence	Remaining implementation
Mains lost / restored	No manual-start latch demonstrated	Fresh start / held-start rejection not implemented
Hub overload recovers	Input fuse auto-resets	Recovery coverage must be resolved without defeating OEM protection
Fan2 /3 /4 fails	Local LEDs; only fan1 RPM forwarded	No independent four-fan protective monitoring claimed
Service / guard removal	No rated isolation assembly selected	Prevent reconnection; verify run-down and inaccessible impellers
PWM/monitoring failure	No stopped-state guarantee	Risk allocation, loss-of-signal test and separate physical protection

Scope to close, not a generic review: (1) determine actual reachable hazards and accept protection/restart requirements; (2) select rated PR1, all supply sensing and DC switching together; (3) establish wire/connector and protection coordination from measured/OEM current/time data; (4) integrate enclosure, protected entries and service isolation; (5) witness actual power restoration, held START, reset, fault recovery and safe-access cases. Existing restart rules are unchanged.

Deliverables included: POWER_COORDINATION.json has 20 load and 36 wire-drop scenarios; scripts/analysis/d01_electrical_closure.py reproduces them. Ten analytic/synthetic tests check arithmetic and that no synthetic result grants energization. Physical prototypes/tests remain 0. No component purchases, live fault tests or invented India stock/prices.

Primary references checked 5 October 2026: ARCTIC P14 Max specification sheet; Noctua NV-PS1 / NA-FC1 / NA-FH1 specifications and NV-PS1 overview; NA-FH1 features. Exact links are in POWER_COORDINATION.json. HSE maintenance guidance is used only for isolation principles, not Indian law or device compliance.

Recommendation: retain the supervised indoor OEM-chain reference. Do not energize this incomplete circuit or bypass PR1. A qualified reviewer must resolve the specific physical/electrical decisions above; the assistant supplies the calculations and drawing evidence, not a false safety approval.